

CROWDCITY AI

A Full-Stack AI-Powered Civic Reporting & Government Services Platform

Candidate Name:	Dhanush Raagav S
Project Focus:	Civic Grievances Redressal & Intelligent Citizen Services Engine
Tech Stack:	HTML5, CSS3, JavaScript, Express.js, Supabase PostgreSQL, Groq AI API
Architecture:	Dual-mode Isolated Module Design (Citizen V1 + Government V2)

Submitted for Professional Technical Interview & HR Review

1. Project Concept & Core Intent

Why We Chose This Project:

In developing nations like India, citizen engagement suffers due to two primary systemic challenges:

- 1. **Civic Grievances:** Disjointed reporting channels, leading to slow response times for issues like potholes, streetlights, and sanitation.
- 2. **Government Welfare Fragmentation:** Access to welfare schemes remains heavily scattered, complex, and paper-based. Eligible citizens are often unaware of their rights or unable to navigate complex qualification checklists.

CrowdCity AI bridges this gap. It acts as an integrated **Civic Grievance & Welfare Application Hub**. It empowers users to report locality issues directly on a map and automatically guides them through matching, checking, and applying for government schemes using advanced natural language algorithms.

2. Complete Technical Stack & Tools Used

Component	Tech Description
Frontend UI	Modern CSS3, HTML5, Vanilla JavaScript (ES6+), FontAwesome Icons
Mapping	Leaflet.js open-source map integration with custom pin overlays
Backend Router	Node.js with Express.js REST API server framework
Database	Supabase PostgreSQL with schema isolation and Row Level Security (RLS)
AI Engine	Groq SDK leveraging Llama-3.3-70b-versatile for sub-second NLP

3. Agentic Development Environment

The project was designed, built, and tested in pair programming collaboration with **Google Antigraivty AI**, a deep coding assistant. The agent facilitated:

- High-fidelity automated code modifications with multi-chunk file updates.
- Deployment and synchronization of localized dictionary assets (English & Tamil).
- Rigorous testing of API response structures, mock fallbacks, and database schema updates.

4. Key Contributions & Implemented Features

Version 1.0: Civic Issue Reporting Ecosystem

- **Issue Filing:** Allowed citizens to submit reports with location details, category tag, and image uploads.
- **Leaflet Live Map:** Placed interactive visual markers representing reported problems across Tamil Nadu cities.
- **Authority Dashboard:** Provided government officers with status assignment workflows (Pending, In Progress, Resolved).

Version 2.0: Unified Government Services Portal

Extends CrowdCity with a fully live, direct citizen utility module comprising 9 main features:

1. **Citizen Services Dashboard:** Central landing page summarizing bookmarks, profile status, and office locators.
2. **Scheme Eligibility Checker:** An interactive profile evaluation wizard applying state eligibility limits.
3. **AI Government Assistant:** A full-page conversational Chatbot clarifying scheme requirements.
4. **Secure Document Wallet:** Securely stores certificates with browser-compliant safe binary Blob rendering.
5. **AI Document Verification:** OCR checks upload clarity, readability, and border boundaries.
6. **AI Form Filling Assistant:** Offers field-by-field guidelines for government application sheets.
7. **Office Locator:** Maps nearby e-Sevai, Taluk, and VAO offices with hours.
8. **Application Tracker:** Organizes submissions and timelines on official portals.
9. **Smart Reminder Center:** Tracks renewal alerts and appointment notifications.

5. Production-Ready System Polish

- **UX Integration:** Created a unified global **Command Palette (Ctrl + K)** for instant navigation across all features.
- **Error Resilience:** Designed custom client pages: 404 (Page Not Found), 500 (Server Error Recovery), and Offline fallback templates.
- **Localization support:** Implemented instant, full-page dynamic translation catalogs for both English and Tamil.